INSTAGRAM LENS

An AI-Powered Intent Layer for a Personalized Experience

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ABOUT INSTAGRAM

One of the world's leading social media platforms at the intersection of social networking, entertainment, and commerce.

Feed Posts

Photos & Videos

Stories

Ephemeral Content

Reels

Short-form Video

DMs

Direct Messaging

Creator Tools

Business & Monetization

Ads System

Advertising Platform

KEY INSIGHTS FROM USER DATA

70%+

of feedback mentions excessive ads

60%+

report addictive behavior patterns

50%+

feel feed is irrelevant to their interests

"90% of my feed is random reels from accounts I don't follow. Where did my friends go?"

"The ads are so frequent they literally break my scroll rhythm"

"I came for photography but now it's just memes and clickbait videos"

PROBLEM DEFINITION

WHO	Instagram's 2B+ daily active users who open the app with different intents at different times — connecting with friends, seeking inspiration, or casually browsing.
WHAT	A one-size-fits-all algorithm that ignores user intent, flooding feeds with irrelevant suggested content and excessive ads — leading to frustration, ad fatigue, and declining content quality perception.
WHY	Users feel trapped: addicted to the content loop but increasingly frustrated by intrusive ads and algorithmically-pushed content they never asked for. This erodes trust and long-term retention.

THE SOLUTION

Instagram Lens

An AI-powered, toggleable Intent Layer that reshapes the UI and algorithm in real-time based on what the user wants right now.

FOCUS

Prioritize friends & close connections

SPARK

Curated inspiration by sub-niche

DISCOVER

Normal mixed discovery feed

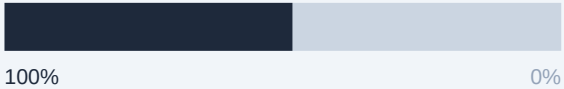
FOCUS MODE

Deep connection with the people who matter most

- Deep-prioritizes friends, close friends, and group activity
- Removes 90% of suggested/algorithmic content
- Time-boxed usage: select duration, then locks for 1 hour
- Promotes intentional, mindful social media use

GREYSCALE MODE

Optional toggle: screen gradually dims and desaturates over 15 minutes of use



Screen locks at 0% brightness

Time Lock Mechanic: After the selected Focus session ends, the mode locks for 1 hour — encouraging users to step away and return with intent.

Greyscale Dimming: If enabled, the screen progressively loses color and brightness over 15 min. At 0%, the app auto-locks — a gentle nudge to stop scrolling.

SPARK MODE

Inspiration, curated by your passions

SPARK STACK

12 sessions

- Explore/Reels style but filtered by specific sub-niches
- User selects a niche: Photography, Cooking, Tech, Fashion, etc.
- 1 ad every 8-10 reels — matched to the selected niche intent
- Promotes discovery within areas the user actually cares about

SESSION STACK TRACKER

A small counter at the top-right of the Spark screen shows how many Spark sessions the user has completed.

Photography	5x
Cooking	4x
Tech	3x

Intent-Based Ads: 1 creative native ad every 8-10 reels, matched to the user's selected niche. Photography lover? See camera gear ads. Cooking fan? See kitchen brand content.

DISCOVER MODE

The classic Instagram experience — familiar and open

- The normal mixed feed: friends + suggested + ads
- Default mode for users who don't actively choose a lens
- Standard algorithmic recommendations
- Creative ads blended naturally into the content flow
- Users can set any mode as default in Settings

Default Setting: Users can pick their preferred default mode in Settings and switch anytime. On app open, a non-intrusive prompt asks which lens to activate.

USER EXPERIENCE FLOW

01
Open App

Non-intrusive Lens selector appears

→

02
Choose Lens

Focus, Spark, or Discover mode

→

03
Experience

UI & algorithm adapt in real-time

→

04
Settings

Set default mode or change anytime

MONETIZATION STRATEGY

Intent-Based Targeting

Ads matched to user's active mode & niche yield significantly higher CTR and conversions

Creative Native Ads

Ads designed as scrollable content — doesn't look like an ad at first glance across ALL modes

Premium Ad Pricing

Fewer ad slots but each carries premium pricing due to higher intent signals and engagement

Spark Mode Ads

1 ad every 8-10 reels, matched to user's selected niche — photography ads for photography lovers

EXPECTED IMPACT

2-3x

↑ CTR

Higher click-through on intent-matched ads

+25%

↑ Retention

Users stay longer with relevant content

+40%

↑ Revenue/Ad

Premium pricing for fewer, better-placed ads

-60%

↓ Ad Fatigue

Fewer complaints, better user sentiment

ASSUMPTIONS & TRADE-OFFS

ASSUMPTIONS

- Users have distinct intents when opening Instagram
- Intent-based targeting increases ad value enough to offset lower volume
- Users will actively engage with the Lens selector
- Creative native ads maintain engagement without feeling intrusive

TRADE-OFFS

- Short-term ad revenue dip from fewer impressions in Focus mode
- Engineering complexity of real-time algorithm switching
- Risk of low adoption if Lens selector feels cumbersome
- Greyscale dimming may reduce engagement metrics short-term

Net Assessment: Long-term gains in retention, user trust, and premium ad pricing outweigh short-term impression volume loss. Intent signals are the future of digital advertising.

THANK YOU

Instagram Lens — Seeing What Matters

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